

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-ad422bd3068fee6e5.jsonl`  
- SHA-256: `83af33912809ce29f18995c9c63755849c6876b817fc152c093aa32576ca582d`  
- Source modified: `2026-07-02T03:02:51+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T03:01:07.318Z)

You are an adversarial verifier in a tech-debt audit. Your default stance: REFUTE. Today is 2026-07-02.

A finder claims the following about the repo at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer:

```
{  
  "title": "No lockfile: runtime closure resolved fresh from floor-only pins on every CI run and image build",  
  "category": "tech-debt",  
  "severity": "P2",  
  "file": "pyproject.toml",  
  "line": 27,  
  "evidence": "pyproject.toml:27-45 and requirements.txt:1-30 use only `>=` floors (fastapi>=0.111.0, numpy>=1.26.4, supervision>=0.21.0, ...); repo root has no uv.lock/constraints/requirements-lock (verified via ls). The nightly license lane (.github/workflows/nightly.yml:165-189) exists precisely because the transitive closure drifts, but nothing makes the closure reproducible.",  
  "why_it_matters": "Any new upstream release (or a hijacked package) flows unreviewed into CI, the serving image, and nightly GPU boxes; it also makes 'CI green' non-reproducible between runs ? the exact drift class the license lane was built to catch, but for correctness/security.",  
  "proposed_fix": "Add a compiled, hash-pinned lock (uv lock or pip-compile --generate-hashes) consumed by CI and deploy/cloudrun/Dockerfile, refreshed via a scheduled bump PR; keep pyproject floors as-is for library consumers.",  
  "effort": "M",  
  "tracked_in": ""  
}
```

Repo root candidates: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer, C:/Users/avidu/Projects/khelsutra-guru/khelsutra, C:/Users/avidu/Projects/khelsutra-guru/sports-data-collector, C:/Users/avidu/Projects/khelsutra-guru/rally-annotator, C:/Users/avidu/Projects/khelsutra-guru/sports-obsreport (pick from the finding's repo field; non-git dirs: 'C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup', C:/Users/avidu/Projects/khelsutra-guru/khelsutra-screencast).

VERIFY against the CURRENT working tree (just pulled to origin master/main): (1) does the cited code actually exist and behave as claimed at file:line? (2) was it already fixed by a recent commit (git log --oneline -15 -- <file>)? (3) is the severity honest - construct the concrete failure scenario or downgrade; (4) is the proposed fix sane and correctly scoped; (5) is it owner-gated per: alpha/serving go-live, Studio P10/P11 corpus-promotion/training, product/strategy/licensing, real cloud spend, counsel ToS/DPA?

HARD RULES: READ-ONLY (no file modifications, no installs, no test-suite runs). gh read-only commands allowed. Your reasoning MUST cite file:line you actually read. CONFIRMED requires you personally saw the defect/debt in current code; if you cannot reproduce the claim from the code, say REFUTED; if a commit fixed it, ALREADY_FIXED.

2. user (2026-07-02T03:01:16.852Z)

```
1      # PEP 621 packaging for the badminton/sports highlight indexer (DEFERRED #7).
2      #
3      # Build backend: hatchling. `backend/` is an intentional NAMESPACE package
4      # (no `backend/__init__.py`; mypy.ini sets explicit_package_bases for the same
5      # reason), so the wheel target packages are listed EXPLICITLY below ? hatchling's
6      # auto-detection would otherwise miss the un-__init__'d `backend` root.
7      #
8      # torch/torchvision are deliberately NOT listed in [project.dependencies]:
9      # their CPU/CUDA wheels need an out-of-band index (--index-url / --extra-index-url),
10     # which PEP 621 cannot express. Install them separately ? see [project.optional-
dependencies]
11     # `gpu` (documentation only) and the README. CI installs CPU torch on its own line.
12     [build-system]
13     requires = ["hatchling"]
14     build-backend = "hatchling.build"
15
16     [project]
17     name = "badminton-highlight-indexer"
18     version = "0.1.0"
19     description = "AI-assisted rally segmentation and highlight indexing for racket sports."
20     readme = "README.md"
21     requires-python = ">=3.10"
22     license = { text = "MIT" }
23     authors = [{ name = "Avi Dullu", email = "avi.dullu@gmail.com" }]
24     keywords = ["badminton", "sports", "video", "highlights", "rally", "segmentation"]
25
26     # Runtime deps mirror requirements.txt, EXCEPT torch/torchvision (see header).
27     dependencies = [
28         "fastapi>=0.111.0",
29         "uvicorn>=0.30.1",
30         "opencv-python>=4.9.0.80",
31         "numpy>=1.26.4",
32         "pydantic>=2.7.1",
33         "jinja2>=3.1.4",
34         "python-dotenv>=1.0.0",
35         "google-genai>=2.4.0",
36         "supervision>=0.21.0",
37         # Used directly on the serving path (backend/features/rally_seq.py: uniform_filter1d
/
38         # maximum_filter1d); previously present only TRANSITIVELY via supervision, so a
supervision
39         # bump or a pruned-transitive freeze could silently break inference. Declare it.
(BSD-3.)
40         "scipy>=1.10",
41         # D13/P12: ByteTrack moved to the standalone `trackers` package (Apache-2.0). Its
42         # PyPI wheel is broken (metadata-only), so it MUST be installed from git ? see
43         # requirements.txt. PEP 621 cannot express a git URL, so it's NOT listed here
44         # (same pattern as torch/torchvision). CI installs it on its own line.
45     ]
46
47     [project.optional-dependencies]
48     # Test + lint/type tooling ? matches the CI lanes (.github/workflows/ci.yml).
49     dev = [
50         "pytest",
51         "pytest-cov",
52         "httpx",          # fastapi TestClient transport
53         "ruff",
54         "mypy",
55         "PyJWT[crypto]>=2.8", # M9 edge-auth: Cf-Access JWT verify (RS256) ? tests + CI
56     ]
57
58     # M9 edge-auth (Cloudflare Access). DEFAULT-OFF: only needed when
security.edge_auth_enabled=True.
```

```

59     # `jwt` is lazy-imported, so the base runtime works without this; install it for
hosted/multi-tenant.
60     auth = ["PyJWT[crypto]>=2.8"]
61
62     # Per-video observability reports (SOFT dependency: the pipeline degrades to a
63     # no-op if absent). Pinned to its git tag per BACKLOG. The private repo is
64     # fetched over SSH; if you lack access, simply omit this group ? reporting
65     # tests skip via pytest.importorskip. For local dev against a sibling checkout:
66     # pip install -e ../sports-obsreport --no-build-isolation
67     reporting = [
68         "obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3",
69     ]
70
71     # `gpu` is DOCUMENTATION ONLY ? it is intentionally empty. torch/torchvision
72     # CUDA wheels require `--index-url`/`--extra-index-url`, which PEP 621 cannot
73     # express, so they are NOT declarable here. Install them out-of-band instead:
74     # # CUDA 12.4 box:
75     # pip install torch>=2.12.0 torchvision>=0.27.0 --index-url
https://download.pytorch.org/whl/cu124
76     # # CPU-only box (also what CI uses):
77     # pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu
78     gpu = []
79
80     # Storage backends (docs/DECOUPLED_COMPUTE/). The default `local` backend needs NONE of
81     # these. Cloud SDKs are imported lazily, so `import backend.storage` works without them;
82     # install the extra for the bucket you use. (boto3's S3 client covers AWS + Cloudflare
R2 +
83     # DO Spaces + MinIO via endpoint_url.) `gdrive://` needs NO extra ? it reads a locally
84     # MOUNTED Google Drive (Google Drive for Desktop) as plain files; see
backend/storage/gdrive.py.
85     storage-s3 = ["boto3>=1.34"]
86     storage-gcs = ["google-cloud-storage>=2.16"]
87
88     [project.scripts]
89     # `indexer --server`, `indexer --help`, etc. -> backend/main.py::main()
90     indexer = "backend.main:main"
91
92     [tool.hatch.metadata]
93     # The `reporting` extra pins obsreport to a git tag (a PEP 508 *direct
94     # reference*); hatchling requires opting in to allow those in metadata.
95     allow-direct-references = true
96
97     [tool.hatch.build.targets.wheel]
98     # `backend` is a namespace package (no __init__.py); list both import roots
99     # explicitly so an editable/wheel install exposes `import backend` and
100     # `import training` regardless of cwd.
101     packages = ["backend", "training"]
102

```

3. user (2026-07-02T03:01:17.437Z)

```

CLAUDE.md
README.md
__pycache__
artifacts
backend
backend.db
config.json
deploy
docs
eval_baselines
indexer.db
mypy.ini
output
pyproject.toml

```

pytest.ini
rally_index.db
requirements.txt
ruff.toml
run_all_tests.py
run_eval.py
run_regression.py
scratch
scripts
tests
tools
training

4. user (2026-07-02T03:01:24.309Z)

```
1      fastapi>=0.111.0
2      uvicorn>=0.30.1
3      opencv-python>=4.9.0.80
4      numpy>=1.26.4
5      pydantic>=2.7.1
6      jinja2>=3.1.4
7      python-dotenv>=1.0.0
8      google-genai>=2.4.0
9      # torch/torchvision are listed here but deliberately NOT in pyproject.toml
10     # [project.dependencies]. Their CPU/CUDA wheels need an out-of-band index
11     # (--index-url / --extra-index-url) that PEP 621 cannot express. Install
12     # them separately matching your hardware:
13     #   NVIDIA GPU: pip install torch torchvision --index-url
https://download.pytorch.org/whl/cu124
14     #   CPU only:   pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
15     # CI installs CPU torch on its own line in .github/workflows/ci.yml.
16     torch>=2.12.0
17     torchvision>=0.27.0
18     supervision>=0.21.0
19     # D13/P12: ByteTrack moved out of `supervision` (removed in 0.30.0) to the standalone
20     # `trackers` package. The PyPI wheel is BROKEN (ships metadata only, no source ? see
21     # https://github.com/roboflow/trackers/issues/474), so install from git until fixed:
22     #   pip install "trackers @ git+https://github.com/roboflow/trackers.git@2.5.0"
23     # (Apache-2.0; depends on supervision >=0.26.1, so it composes cleanly.)
24     # NOTE: roboflow/trackers uses bare version tags (2.5.0), NOT v-prefixed.
25     # PIN-DUPLICATION (mirrors the torch carve-out): this exact ref is ALSO hard-coded in
26     # .github/workflows/ci.yml + nightly.yml (full-suite lanes) and
deploy/digitalocean/bootstrap.sh
27     # + deploy/cloudrun/Dockerfile ? edit ALL of them in lockstep on a bump, or the
lanes/boxes
28     # silently resolve different versions.
29     trackers @ git+https://github.com/roboflow/trackers.git@2.5.0
30     scipy>=1.10
31
32     # Per-video observability reports (SOFT dependency: the pipeline degrades to a
33     # no-op if absent). Shared, permissively-licensed report toolkit. Pin a tag once
34     # the private repo is published, e.g.:
35     #   obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3
36     # For local development install it editable from a sibling checkout:
37     #   pip install -e ../sports-obsreport --no-build-isolation
38
```

5. user (2026-07-02T03:01:26.561Z)

3624a8d feat(D13/P12): migrate ByteTrack to standalone trackers package (#386)
627ff68 docs: explain torch split between requirements.txt and pyproject.toml (Q5) (#359)
e8bad40 feat(pkg): P0d ? declare scipy (undeclared serving dep) + license-scan & built-wheel CI
lanes (#297)
aac7633 feat(serving): M9-auth ? Cloudflare Access edge-auth + per-tenant owner_id (default-OFF)

```
(#290)
38c2d0f feat(storage): gdrive:// reads a locally MOUNTED Google Drive (no API) (#258)
6878d45 feat(storage): pluggable StorageBackend ? local + S3/GCS first-class buckets (#249)
c143f40 feat(packaging): PEP 621 pyproject + editable install + entry point (DEFERRED #7)
b8904dc chore(reporting): per-tool customer_verdict + bump obsreport pin to v0.1.3
513c754 feat(reporting): always-on per-video observability report (indexer)
d19557e feat(indexer): replace AGPL ultralytics YOLO with permissive torchvision + ByteTrack
9c0e3c5 Update PyTorch installation for GPU & add configurable Gemini model
3da56b5 Initial commit: Complete Badminton Highlight & Rally Indexer with H.265 sequential
optimizations and max-frames limit
---
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)
---
```

6. user (2026-07-02T03:01:26.665Z)

badminton-highlight-indexer\artifacts\cloudrun_lightup\output\sports_indexer.db.lock

7. user (2026-07-02T03:01:27.780Z)

badminton-highlight-indexer\training\requirements-train.txt
badminton-highlight-indexer\requirements.txt

8. user (2026-07-02T03:01:35.493Z)

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/.github/workflows/:
ci.yml
nightly.yml

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/deploy/:
cloudrun
digitalocean
gcp

9. user (2026-07-02T03:01:36.382Z)

```
140
141     # Offline quality guard: the committed golden regression fixtures
142     # (docs/GOLDEN_REGRESSION_FIXTURES.md) ? synthetic + de-attributed real
143     # trajectory fixtures, NO video / GPU / DB. Mirrors the offline replay in
144     # ci.yml::golden-crossos (single-OS here; ci.yml already covers cross-OS on
145     # every PR ? nightly is about drift-over-time on the default branch). The
146     # CONTROL replay needs only numpy.
147     golden-fixtures:
148       name: Nightly · Golden fixtures regression (offline quality guard)
149       runs-on: ubuntu-latest
150       timeout-minutes: 15
151       steps:
152         - uses: actions/checkout@v6
153         - uses: actions/setup-python@v6
154           with:
155             python-version: "3.13"
156         - name: Install minimal deps (no GPU stack ? the CONTROL replay needs only numpy)
157           run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
genai numpy
158         - name: Golden-fixtures characterization (offline)
159           run: |
160             python -m backend.eval.golden_fixtures --check
161             python -m pytest tests/test_golden_fixtures.py
tests/test_golden_real_fixtures.py -q
162
163     # Commercial-clean licensing guardrail (MIT/Apache/BSD only) ? mirrors
```

```

164 # ci.yml::license-scan. Scans the RUNTIME dependency closure and fails on any
165 # copyleft (GPL/AGPL/LGPL/SSPL). This is exactly the transitive-dep drift the
166 # header cites as a reason the nightly exists ? a new sub-dep can flip a
167 # license between PRs without touching our code.
168 license-scan:
169   name: Nightly · Dependency license scan (no copyleft in runtime deps)
170   runs-on: ubuntu-latest
171   timeout-minutes: 10
172   steps:
173     - uses: actions/checkout@v6
174     - uses: actions/setup-python@v6
175       with:
176         python-version: "3.13"
177     - name: Install runtime deps + pip-licenses
178       run: |
179         python -m pip install .
180         python -m pip install pip-licenses
181     - name: Report dependency licenses
182       run: python -m piplicenses --format=markdown --with-urls --order=license
183     - name: Fail on any copyleft (GPL/AGPL/LGPL/SSPL) in the runtime closure
184       run: |
185         if python -m piplicenses --format=plain | grep -iE "GPL|SSPL"; then
186           echo "::error::A runtime dependency carries a copyleft (GPL/AGPL/LGPL/SSPL)
license ? the guardrail is MIT/Apache/BSD only. Review the report above."
187           exit 1
188         fi
189       echo "OK: no GPL/AGPL/LGPL/SSPL in the runtime dependency closure."
190
191 # Built-wheel smoke (install + launch) ? mirrors ci.yml::build-artifact. Builds
192 # the wheel, installs it NON-editable from a non-repo cwd, and launches the
193 # console script. Catches packaging regressions the source suite can't.
194 build-artifact:
195   name: Nightly · Built wheel smoke (install + launch)
196   runs-on: ubuntu-latest
197   timeout-minutes: 20
198   steps:
199     - uses: actions/checkout@v6
200     - uses: actions/setup-python@v6
201       with:
202         python-version: "3.13"
203     - name: System libs for opencv-python
204       run: |
205         # Drop the runner's pre-installed Microsoft apt repo (azure-cli): it
periodically
206         # 403s on InRelease and aborts `apt-get update` under the `-e` shell, failing
the
207         # build on an unrelated repo we don't use. libgl1/libglib2.0 are in Ubuntu's
archive.
208         sudo rm -f /etc/apt/sources.list.d/*microsoft* /etc/apt/sources.list.d/*azure*
209         sudo apt-get update
210         sudo apt-get install -y libgl1 libglib2.0-0t64 || sudo apt-get install -y
libgl1 libglib2.0-0
211     - name: Build the wheel
212       run: |
213         python -m pip install build
214         python -m build --wheel
215     - name: Install the built wheel + CPU torch (out-of-band index)
216       run: |
217         python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
218         python -m pip install dist/*.whl
219     - name: Launch the installed console script from a non-repo dir

```

```

30:      run: python -m pip install ruff
79:      run: python -m pip install mypy pydantic fastapi uvicorn python-dotenv google-genai
numpy
97:      run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-genai
123:     run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
132:     run: python -m pip install "trackers @
git+https://github.com/roboflow/trackers.git@2.5.0"
136:     run: python -m pip install -e .[dev]
167:     run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
genai numpy
177:  # RUNTIME dependency closure (`pip install .` ? no torch/dev) and fails on any copyleft
195:     python -m pip install .
196:     python -m pip install pip-licenses
230:     python -m pip install build
234:     python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
235:     python -m pip install dist/*.whl

```

11. user (2026-07-02T03:01:46.134Z)

```

badminton-highlight-indexer\deploy\gcp\README.md:181:. .venv/bin/activate && pip install -e
.[storage-gcs] # GCS backend
badminton-highlight-indexer\deploy\cloudrun\Dockerfile:43: &&
"$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
badminton-highlight-indexer\deploy\cloudrun\Dockerfile:45: && ( [ -f
"$WASB_REPO_DIR/requirements.txt" ] && "$CONDA_DIR/envs/wasb/bin/python" -m pip install -q -r
"$WASB_REPO_DIR/requirements.txt" || true ) \
badminton-highlight-indexer\deploy\cloudrun\Dockerfile:46: &&
"$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
badminton-highlight-indexer\deploy\cloudrun\Dockerfile:59:# roboflow/trackers#474) so it's NOT
pulled by `pip install .`; install it out-of-band before
badminton-highlight-indexer\deploy\cloudrun\Dockerfile:61:RUN python3 -m pip install --upgrade
pip \
badminton-highlight-indexer\deploy\cloudrun\Dockerfile:62: && python3 -m pip install
"trackers @ git+https://github.com/roboflow/trackers.git@2.5.0" \
badminton-highlight-indexer\deploy\cloudrun\Dockerfile:63: && python3 -m pip install
".[storage-gcs]"
badminton-highlight-indexer\deploy\digitalocean\gpu_setup.sh:43:python -m pip install -q
torch==1.11.0+cu113 torchvision==0.12.0+cu113 \
badminton-highlight-indexer\deploy\digitalocean\gpu_setup.sh:45:[ -f
"$WASB_REPO/requirements.txt" ] && python -m pip install -q -r "$WASB_REPO/requirements.txt" ||
true
badminton-highlight-indexer\deploy\digitalocean\gpu_setup.sh:46:# WASB-SBDT's requirements.txt
is INCOMPLETE: src/ imports hydra, pandas, etc. that it omits, so
badminton-highlight-indexer\deploy\digitalocean\gpu_setup.sh:49:python -m pip install -q hydra-
core omegaconf pandas opencv-python-headless pillow scipy tqdm pyyaml matplotlib "numpy<1.24" ||
true
badminton-highlight-indexer\deploy\gcp\nightly_regression\run_on_vm.sh:90:pip install -e
'.[storage-gcs]
badminton-highlight-indexer\deploy\digitalocean\bootstrap.sh:29:python -m pip install --upgrade
pip
badminton-highlight-indexer\deploy\digitalocean\bootstrap.sh:37: pip install torch torchvision
--index-url https://download.pytorch.org/whl/cu124
badminton-highlight-indexer\deploy\digitalocean\bootstrap.sh:39: pip install torch torchvision
--index-url https://download.pytorch.org/whl/cpu
badminton-highlight-indexer\deploy\digitalocean\bootstrap.sh:46:pip install "trackers @
git+https://github.com/roboflow/trackers.git@2.5.0"
badminton-highlight-indexer\deploy\digitalocean\bootstrap.sh:49:pip install -e .[dev]

```

12. user (2026-07-02T03:02:01.239Z)

```

badminton-highlight-indexer\backend\ui\THIRD_PARTY_NOTICES.md:8:see its package
manifest/lockfile for the authoritative per-package licenses.
badminton-highlight-indexer\docs\archives\TEST_RUN_SUMMARY.md:18:During the raw test execution

```

on the 11.5 GB video stream, the system hit three distinct, high-impact constraints. We fully diagnosed, optimized, and hardened the pipelines to address them:

badminton-highlight-indexer\docs\HOSTING_PLAN.md:4:**Date:** 2026-06-10 . **Constraints:** budget \$10?25/mo . GCP credits available . DigitalOcean

badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:49:>

[`DEFERRED\_HARDENING\_PLAN.md`](DEFERRED_HARDENING_PLAN.md). Commercial constraints come

badminton-highlight-indexer\docs\archives\past_projects\DECOUPLED_COMPUTE\01-EXISTING-PLANS-AND-GAP.md:123:**Three further constraints sit beside the gap** (detailed in [06](06-RISKS-AND-OPEN-QUESTIONS.md)): the

badminton-highlight-indexer\docs\RALLY_QUALITY_RESEARCH.md:15:constraints + cross-links.

badminton-highlight-indexer\docs\RALLY_QUALITY_RESEARCH.md:570:## 8. Constraints & guardrails (non-negotiable) + cross-links

13. user (2026-07-02T03:02:50.976Z)

Structured output provided successfully